

# IDEAS

Sheet #1

Author Clarence Ang Jule

Date 15/10/2025

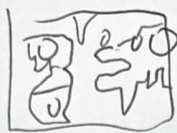
Digitalisation

Socioeconomic Factors

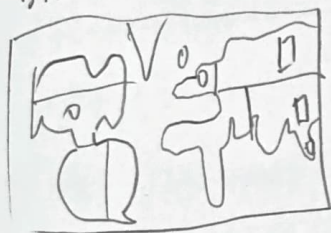
World

Datasets

- Research patents
- ICT infrastructure
- % of internet users
- % of people who own computers
- % in university
- % of researchers in country
- 5G connectivity
- Internet speed

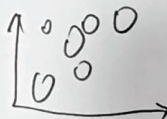
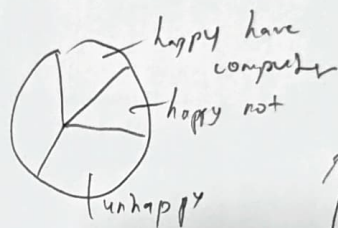


Bivariate



Datasets

- GDP
- GDP per capita
- Well-being indicator
- HDI
- Inequality
- Life satisfaction index



FILTER

- Choropleth map to show ICT
- % of internet users
- HDI map
- Inequality
- Well-being indicators
- Bar chart
- map

CATEGORIZE

Digitalisation

- Use map to show infrastructure
- Use line chart to show growth over time
- Use pie chart to show set

Socioeconomic

- year is important to distinguish time
- Can use line too to see well-being
- map which places are happier

REFINE

- Choropleth map
- Line chart
- Bivariate map
- Bar chart
- Line chart from  $y=0$  to show growth
- Bubble chart over time for improvement
- Map with bubble to show the growth

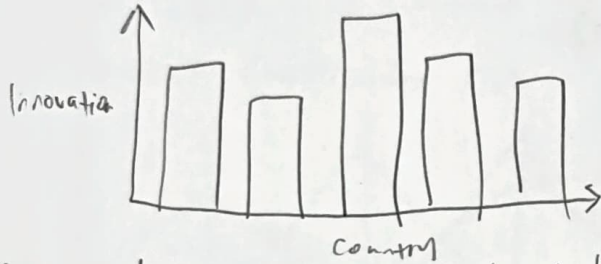
# LAYOUT

Full

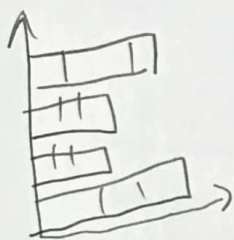
bivariate choropleth map



Bar chart - Top 10 innovators



Bar chart - Top 10 most productive



Pie chart - Technology sector

## FOCUS

- Bivariate choropleth map
- Variable A: GDP/per capita: Hue
- Variable B: (Internet usage): Saturation



- Low Hue, Low Saturation: Poor & lack of tech
- Low Hue, High Saturation: Poor & high connectivity
- High hue, low saturation: Rich & low connectivity
- High hue, High saturation: Rich & high connectivity

Title: The Global Divide

Author: Clara Ang Jia Ler  
(33307873)

Sheet: 2

Date: 15/10/2025

Task: To show economic and digital landscape

## OPERATIONS

- Single select year slider across the sheet

Select year ~~2020~~ 2025



- Hovering displays tooltip
- Slider reacts to changing the map

## DISCUSSION

### Advantages

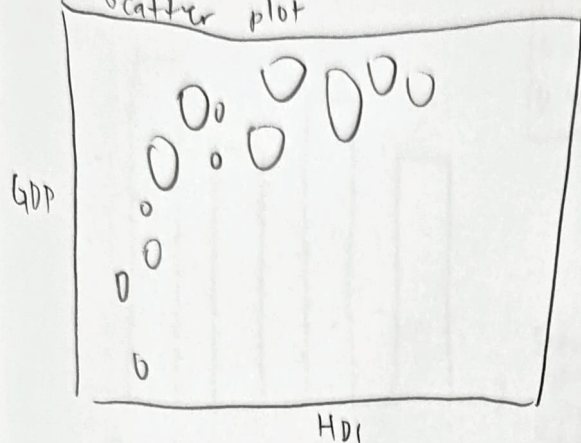
- High information density
- Clear, focused narrative
- Great analysis tool

### Disadvantages

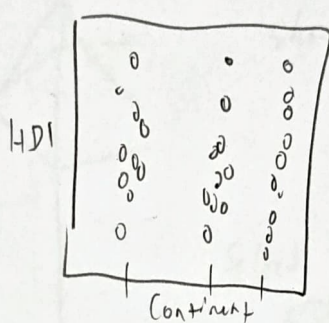
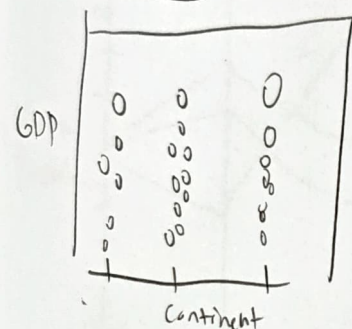
- High cognitive load
- Colorblind people may have issues
- Ranking instability: The bar chart may keep moving around over time



## Animated bubble chart - scatter plot



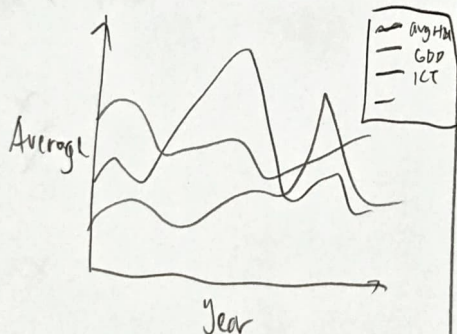
## Categorical



## Line chart



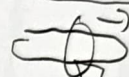
## Average trend line chart



Title: The Core Relationship  
 Author: Clarence Ang Jia Ler  
 Sheet: 3  
 Date: 15/10/2025  
 Topic: To show global change

## OPERATIONS

- Time animation



select year

The year will show changes in the bubbles

- Selected country when user selects a country the line will show



Select country:   
 this shows trend

- If user selects 'ALL' the average trend will appear

## DISCUSSION

### Advantages

- Interesting/Engaging chart
- Storytelling value
- Micro scale to Macro Scale
- High data density

### Disadvantages

- Overlapping
- Color blind people may find it difficult
- Correlation  $\neq$  Causation

## FOCUS

- The primary scatter plot will be the main focus, as it shows core correlation
- The categorical shows inequality
- The select country trend shows the trend that a specific country follows
- The average trend gives users the big picture

## Bivariate Bar Chart



10%  
20%  
30%

Y-axis - Country

X-axis - R&D per million

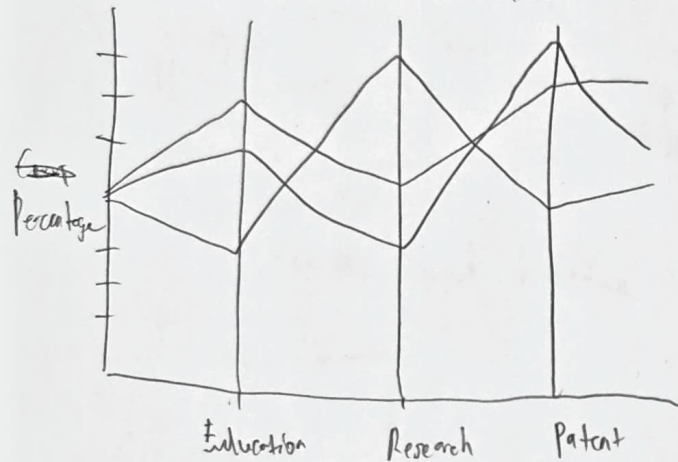
Color - Patent apps/million

Select year



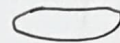
- This shows the user to compare patent & R&D

## Parallel Coordinates Plot

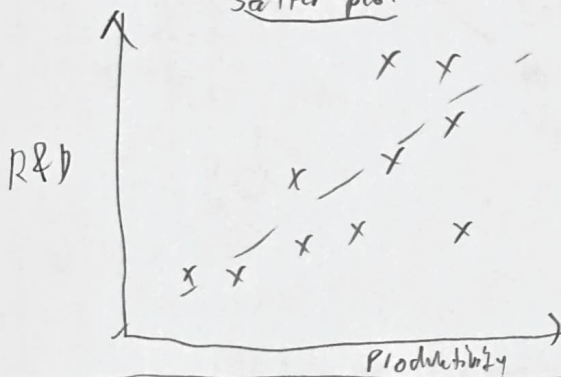


- Shows the pipeline

Select Country



## Scatter plot



Title: The Engine of Growth  
Author: Claudia Argy Jialar

Date: 15/10/2025

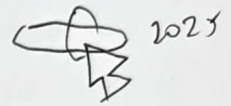
Sheet: 4

Task:

## OPERATIONS

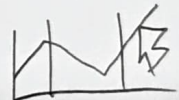
### 1. Coordinated Filtering

- Slider for user to choose the year

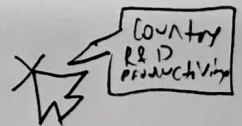


### 2. Highlighting

- User clicks on one line then it highlights it overshadowing the others



### 3. Tooltip



## FOCUS Bivariate Bar Chart

- Focuses users on patents
- Color is mapped to a sequential scale
- Long, Dark bar is an innovation powerhouse
- Short dark bar is few researchers but more patents

## Scatter plot

- Shows whether increased productivity leads to more R&D
- Uses regression line to determine correlation

## Parallel coordinates

- Shows the pipeline of the amount of R&D for each country

## DISCUSSION

### Advantages

- Strong narrative
- Brings user closer to the story

### Disadvantages

- Time Lag
- Data sparsity
- Hard to filter with missing info



# LAYOUT

## Introduction



Sets the scene to allow users to understand the situation

Select year

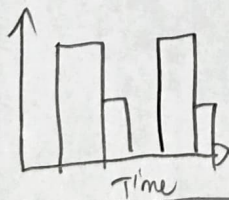
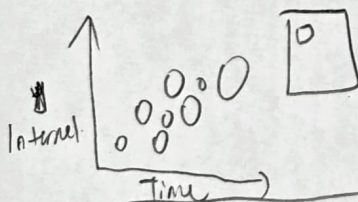
Combine and Refine



Combines data to see if there's any connection



Wealth or Prosperity?



## FOCUS

- The map allows user to look at the socioeconomic scenario of the world
- Then the information is combined to give users a good comparison point
- The final charts use Year to see if any data came before another.

Title: Socioeconomic Outcome & Digital prosperity

Author: Clavencia Any Jia Lw (33207873)

Date: 15/10/2025

Sheet: 5

Task: Realisation design

## Estimation

Maps: 4-5 hours

Bar chart: 1 hour

Scatter: 2 hour

Bubble: 3 hours

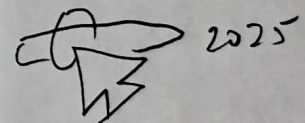
## Dependencies

- Python pandas for preprocessing

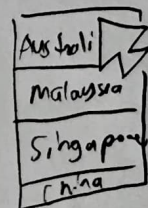
- Vega-Lite for graphs

## Operation

- Year selector



- Country Selector



- Tooltip

